

TLH Sleeve Architecture v2

TLH SLEEVE — FULL ARCHITECTURE v2

Deployment, harvest machinery, AI-drawdown research program, and the hedge decision

SignalOS desk · 2026-08-28 · supersedes TLH_SLEEVE_ARCHITECTURE_20260826 (v1) and incorporates TLH_HEDGE_PROPOSALS_20260828 + AI_BREAK_ENSEMBLE_FINDINGS (2026-07-30)

ABSTRACT

The problem. A ~\$5.0M zero-basis payment lands in September. Roughly \$1.855M of it is already the government's; the remaining ~\$3.2M must be deployed so that enough capital losses are *realized by December 31, 2026* to offset the gain at a ~37.1% marginal rate — inside a household that already carries \$9.2M of externally-managed levered tax-harvesting (Parametric), holds ~26% of its wealth in the AI complex, and expects further inflows.

The design, in one paragraph. The money deploys into a separate rule-driven account (BETA) that can never hold a courted name: ~\$1.9M goes directly into 32 pre-ruled single-name slots (single names disperse — dispersion is what produces harvestable losers before year-end) and ~\$1.3M into a two-fund index completion sleeve whose job is beta, not losses. Two rotation rules convert index dips and court gates into harvests; every name carries a pre-assigned wash-safe replacement partner (the pair map, with November 28 as the last doubling-up date); and a small code layer — pair validation, cross-account wash checks, a constructive-sale screen that already rejected the obvious short candidates — makes the tax rails machine-enforced rather than remembered. Order-integrity sentinels, band-tripwire discipline, and a live status page provide the operational floor, each guard built from a failure the desk actually observed.

The risk, and the hedge derived from first principles. The desk's seven-model ensemble puts the probability of an AI-complex credit event by end-2027 at a frozen 0.45, and — more usefully — predicts its *sequence*: credit breaks first (neoclouds, then Oracle, then the credit channel), equity multiples last. Because the household already owns three layers of implicit protection (harvest convexity worth ~37 cents per dollar of drawdown, an anti-correlated tax reserve, and tripwires that fire before the equity leg), and because a net buyer should never sell its melt-up branch, the previously-ratified call-collar is withdrawn. The recommendation is a **two-layer hedge matched to the information structure**: a small standing put-spread (~10–17% of the AI overlap, ~\$10–18k/yr) covering only the no-warning branch, plus a pre-registered escalation to ~50% of the overlap — executed within five sessions of any named gate firing — accepting crisis volatility as the known price of certainty. Cheaper than the collar when nothing happens, larger when it matters, upside never sold.

What awaits the principal: ratify the two-layer hedge and its doctrine amendment (protection phases with deployment *and information*, not calendar); rule on the bounded dispersion sleeve; set the ALPHA-migration gates; complete the BETA activation checklist at account approval. Nothing stages until the money lands.

PART I — THE SLEEVE (deployment + harvest machinery)

1. Objective and binding constraints

Realize enough capital losses by **2026-12-31** to offset the ~\$5.0M zero-basis gain landing in September (~37.1% combined marginal → each harvested dollar saves ~\$0.37 *this year*; 2027 harvests are only carryforwards) — without ever letting the harvest machine contaminate the judgment machine.

#	Constraint	Design consequence
C1	Losses realize by 12/31/26	slots-first (dispersion now); index residual accepts ~zero 2026 harvest
C2	Tax reserve ≈ \$1.855M (true-up at landing)	bills/SGOV day 0; deployable ≈ \$3.2M
C3	Household wash surface = Parametric singles + GOOGL + BETA basket	completion-set construction + pre-rotation sweeps + harvest_engine checks
C4	Household AI complex ≈ \$5.18M ≈ 26% of household (Parametric 38% internal)	no QQQ; broad index only; hedge sized to the OVERLAP (Part III)
C5	Judgment-vs-rule boundary (principal-ratified)	two accounts; courted names structurally barred from BETA
C6	Ops maturity	BETA has NO automated order paths — staged CSVs, principal clicks

2. Account architecture

ALPHA (U24485312) = the court book: every courted name, envelopes, options templates — judgment, gated by courts. BETA (U28113440, pending) = the \$5M: 32 pre-ruled slots + index completion residual + hedge + reserve — rule, no courts. Parametric (\$9.19M, 131/31) = the external levered TLH engine (~\$105k/mo realized losses, -\$425k banked YTD). The wash perimeter is household-wide; `beta_basket_generator.py` makes BETA non-colliding **by construction** (index minus Parametric bundle minus ALPHA book minus GOOGL). `require_alpha()` pins all four automated placement paths to ALPHA; the Error-435 incident is the reason.

3. Deployment lanes (amended 8/21: SLOTS FIRST)

1. **Weeks 1–3:** ~\$1.8–1.9M directly into the 32 pre-ruled slots (lots 2–3 of the 8/02 map) — staged ladders, envelope-chunked where thin. Every week earlier = more dispersion before 12/31.
2. **Index residual ~\$1.2–1.4M** same week, two non-identical broad funds (VTI+SPLG class) so future index-level harvests swap wash-free. Job = beta, not losses.
3. **Completion criterion:** whatever index remains unrotated at month 12 (~\$0.5M floor) IS the permanent completion sleeve.

4. Harvest machinery

- **Rotation rules:** R1 (harvest-driven — index loss $\geq$ \$5k or a $\leq -1.5\%$ day $\rightarrow$ sell high-basis index lots, buy slot names at bands) and R2 (gate-driven — court gates outrank harvest timing).
- **Harvest-pair map** (33 pairs on file, validated 8/26 against the live wash surface): every slot name has a designated non-identical partner for the 31-day window. *No pair, no harvest.* **Nov 28 = last doubling-up date** (buy the partner 31 days before selling the loser).
- **desk/harvest_engine.py (built 8/26):** Phase 0/1 live — pair-map validation vs the live surface, per-name wash checks (PARAMETRIC_SELF protocol), the **\$1259 constructive-sale short screen** (first run rejected NVDA/AAPL/UNH/DE as short candidates — shorting a name the household holds appreciated force-realizes Parametric's gains: the anti-TLH trap, now machine-enforced), and the R1 staged-basket stub that goes live at landing. Phase 2 (a \$150–300k completion-set short-clip pilot) and Phase 3 (closed-loop) remain principal-gated.
- **Proposed, unratified:** the bounded diligenced-dispersion sleeve (\$750k–1M long-only of FLAT-ruled names passing three filters: liquid, not RP_TAINTED, no live gate in the harvest horizon) — est. 2–3 $\times$ harvest yield per dollar vs the index residual.

5. Operating infrastructure (all observed-failure-driven, none hypothetical)

Order sentinel (30-min: zombie cancels, PendingCancel SLA, cancel-by enforcement, gateway 2-strike alarm, runner liveness) · order-intent ledger (every directive machine-checked) · band watcher w/ live-blotter coverage + 48h touch-disposition invariant · consistency guards (pack dates, drain stalls, **rubber-stamp/artifact guard**) · §BANDS-OVER-RESTING-GTC (tripwires + native alerts over resting GTCs wherever a cancel-by would exist) · pipeline status UI (/status). Cadences: weekly bundle $\rightarrow$ basket regen; monthly migration/hedge-phase/reserve review + MR-scorecard; quarterly AI-break re-freeze (Part II) gating hedge tranche 3.

PART II — THE AI-DRAWDOWN RESEARCH PROGRAM

Why a tax-harvesting document carries a bubble model: C4. The household is ~26% AI-complex, the sleeve adds index beta on top, and the harvest engine's own convexity is a drawdown instrument. The hedge decision (Part III) is priced off this program's frozen number.

6. The seven-model AI-break ensemble (built 7/30; findings doc + runnable models in desk/models/)

Prompted by the “Second Derivative” thesis (AI buildout = credit-financed real-estate cycle that breaks on capex *deceleration* — the 2008 mechanism). Seven independent epistemologies, read for their disagreement:

Member	Vote	One-line reading
Mechanism (refi wall + gates + 2008 clock)	AMBER 0.30	wall binds 2028 only in bust regime; Gate A half-lit
LPPL / Sornette critical-point	RED 0.86*	super-exponential fits on all 3 series; t_c ≈ Dec-2026
Credit-vs-equity basis (the 2007 ABX signature)	GREEN	no divergence <i>because periphery equity is already soft</i> (eq-wt basket -1.6%/6m vs SOXX +40%)
Reference class (8 capex manias)	AMBER 0.46	telecom = closest analog, broke 1-4q post-peak because vendor financing — intensity today HIGH
Minsky financing-stage	RED	AMZN 1.02 / ORCL 1.74 capex-to-OCF = Ponzi-stage ; \$197B trailing net debt issuance
Behavior tells	AMBER	CRWV insiders 10× intensity, 82 sellers — FIRED
Network contagion (Eisenberg-Noe)	structural	invariant order: neoclouds seize first (even at 50% shock) → ORCL secondary-defaults → credit channel → megacap margins → multiples

Frozen: AI-BREAK | 2027-12-31 @ 0.45 (median 0.46, range 0.30–0.86); companion ~0.72 by end-2028. “Credit event” = neocloud default/seizure, take-or-pay impairment, GPU-ABS downgrade cascade, or RPO counterparty reprice — **not an equity drawdown**; equity multiples are the LAST step of the predicted sequence, which is what makes the sequence tradeable.

Convergent findings: CRWV is the canary in three unrelated datasets (Merton DD 1.6, insider 10×, first node to breach in every clearing scenario). ORCL is the concentrated hyperscaler (Ponzi financing + worst hit-to-buffer + equity already -26% while the index rose). The complex is **already repricing beneath the index surface** — the equal-weight periphery is negative while cap-weight melts up, historically the *late* configuration.

Gates and watch order: CRWV credit/insiders (firing) → OpenAI IPO terms (Gate A: pricing <\$852B or slip past Q1-27) → the **Nash flip** (first capex guide-down the tape *rewards*; while all five hyperscalers raise guides, near-term P stays capped). Kill condition for the whole frame: lab revenue becoming independent end-demand rather than recycled capex. **Standing instrumentation:** ai_capex_watch (weekday: realized-accel clock + guide event studies); ensemble re-run quarterly, after hyperscaler prints, and on any gate event — **the quarterly re-freeze is a live input to Part III.**

7. The rest of the AI-drawdown book (the same graph, other edges)

- **WAL NDFI kill-class:** the bank edge of the contagion graph — WAL lends to levered funds; the break transmits through fund-finance. Human-observer channel carries the only sensor (NAV-facility/sub-line tightening chatter); a pattern of observations legitimately moves the ensemble p at re-freeze. Position instrumented (650 sh, kill-triggers armed).
- **AI-crash-insulated CA munis:** the fiscal downstream channel — the muni work screens CA districts for AI-payroll/property-tax sensitivity so the muni sleeve is the *insulated* cohort, not the exposed one (AI_CRASH_INSULATED_CA_MUNI work, muni stack).
- **CRWV canary + CIFR/IREN gates:** the desk's neocloud exposure is gated, never resting — CIFR reopens only $\leq \$12.50$ w/ delivery verified (hiring-confirmed 8/26); the first hyperscaler JV is the kill variable.
- **Software-sleeve displacement watch:** the seat-erosion question (NOW/SAP/MNDY/HUBS + the DOCU IAM/LLM-substitution disclosure) is the *revenue-side* AI risk — tracked per-name in courts, human-observable at renewals.
- **Household quantification:** AI complex \$3.52M inside Parametric (38% of its NAV; NVDA \$575k, AAPL \$561k, MSFT \$399k, AMZN \$295k, GOOGL+GOOG \$485k, AVGO \$257k) + ~\$700k external GOOGL + VTSAX tech weight $\approx$ **\$5.18M on ~\$19M** (26%). A -30% AI drawdown $\approx$ -\$1.55M household.

8. How the AI work and the TLH sleeve interlock (the design's spine)

1. **The hedge is priced off the frozen number** — tranche 3 exists only while the quarterly re-freeze holds ≥ 0.45 (Part III).
2. **The harvest is the second hedge:** a drawdown *feeds the product* — $\sim 37\text{¢}/\$$ of decline in tax value against this year's gain. The two mechanisms split the loss curve: options cover the first leg, the tax code monetizes the rest.
3. **The reserve is anti-correlated by construction** (bills; the gain it reserves against shrinks in the same states).
4. **The sequence is the tripwire ladder:** because the ensemble predicts credit-before-equity, the desk's earliest signals (CRWV credit, neocloud seizure, Gate A) fire *before* the index drawdown the hedge covers — re-freezes can raise the hedge before the equity leg moves.

PART III — THE HEDGE DECISION (derived from first principles; prior ratification set aside at principal direction 8/28)

The full derivation lives in TLH_HEDGE_PROPOSALS_20260828 (v2). Summary of the chain: **Risk:** $5.18M_{\text{householdAIcomplex}}; \text{frozen}P(\text{crediteventbyend} - 27) = 0.45;$
*theequitylegistheLASTstepofthepredictedsequence. * *Pre – existingprotection : * **

harvestconvexity(37cents/), reserve anti-correlation, an alpha book the courts already kept light on AI, and — the underused asset — the sequence itself, whose tripwires fire before the equity leg. **Net-buyer test:** the household expects further inflows; any call leg sells the branch it most wants — caps fail first principles regardless of their carry savings.

RECOMMENDATION — the two-layer hedge, matched to the information structure: -

Layer 1 (standing, small): Dec-27 XND put spreads 10/30% OTM, 15–25 spreads (~10–17% of overlap), ~\$10–18k/yr — insurance for the ONE branch the models admit they would miss (no warning). No call leg. - **Layer 2 (pre-registered, bought only on evidence):** on Gate A / a CRWV credit event / the Nash flip / a re-freeze ≥ 0.55 — escalate within 5 sessions to ~50% of overlap in plain puts, accepting crisis IV as the known price of certainty. Decision made now; execution waits for the signal. - **Layer 0 (owned already):** the tax code and the reserve carry the deep grind; Nov-28 doubling-up is part of the hedge. - **Wind-down:** re-freeze < 0.35 two consecutive quarters -> Layer 1 non-renews. - **Deployment floor:** no hedge notional before its capital lands — protection follows capital AND evidence, never calendar.

Doctrine consequence (requires ratification): conditioned-insurance amends from phase-with-deployment-% to phase-with-deployment-AND-information; the 8/21 collar is withdrawn as default (its call leg fails the net-buyer test; its full standing size pays for warning the ensemble already supplies). Expected cost: ~\$13–24k in the modal no-gate path vs the collar's ~\$33–66k path-independent — cheaper when nothing happens, bigger when it matters, upside never sold.

PART IV — THE ALPHA SLEEVE: PURPOSE, ROTATION POSTURE, AND THE DISASTER BOOK

12. What the alpha sleeve is FOR (four purposes, in priority order)

1. **A deliberate anti-rotation fund.** The founding directive (7/22): *“build our own hedge fund to work against sector rotation.”* The book is a ~90-position, 13-axis multi-sector fundamental long book constructed so that no single rotation carries it — measured, stressed, and construction-planned by `desk/rotation_fund.py`.
2. **The knowledge factory.** Every court feeds detectors into the knowledge graph; the KG, the Brier ledger, and the doctrine ARE the moat (the meta-thesis). The P&L is the grade, not the product.
3. **The edge barbell on the TLH core** — capacity-constrained edges (Japan micro, frontier GDRs, court-gated smallcaps) that big money structurally cannot reach, sized so the ceiling is the feature.
4. **The seed track record** for the first outside-capital product (Antifragile Carry — Part IV §14).

13. How much does it hedge sector rotation? (engine read, 8/28 — marks partially stale, directional)

Diversification: **10.6 effective axes / 21.2 effective names** across the mapped core. Six rotation stresses on the deployed book:

Rotation scenario	Unhedged	With designed IGV overlay	Worst axis
Risk-off (cyclical→defensive)	−11%	−10%	software_saas
Rate spike	−5%	−4%	software_saas
AI-capex derate	−5%	−3%	software_saas
Growth→value	−3%	−2%	software_saas
Software→energy	−1%	−0%	software_saas
US→intl / dollar down	~0%	+0%	software_saas

The honest answer in three sentences. Against *rotation proper* — money moving between sectors/styles — the book is well-built: growth→value costs ~3%, software→energy ~1%, US→international ~zero, because 13 axes catch what each other shed. The two residuals are (a) the **correlated software/IT block (~21% of deployed — the worst axis in all six scenarios)**, for which the designed-but-unstaged IGV short envelopes (368+69 IGV ≈ \$39k) exist awaiting principal staging, and (b) **DFIN at 11%** single-name. And the −11% risk-off row is NOT rotation — it is market beta, which no amount of sector construction hedges; that tail belongs to Part III's XND layers, which is exactly why the hedge decision lives in this document. **Self-cure:** both concentrations are relative to deployed capital; the construction plan routes the ~\$645k of remaining dry powder to the under-weight intended/idio axes (real-asset ballast, Japan value, EM financials, insurance, energy, healthcare) and freezes the software block — deployment fixes concentration without trimming conviction.

14. The disaster book (The Desk's disaster-planning corpus, unified here)

The desk's tail work spans four layers, each with its own instrument:

Layer	Threat	Work product	Instrument / status
Axis	sector rotation	rotation_fund 3-stage engine (measure/hedge/construct)	live; IGV envelopes designed, unstaged
Market	beta drawdown / AI break	7-model AI-break ensemble (Part II), frozen 0.45, credit-first sequence	two-layer XND hedge (Part III, awaiting ratification); harvest convexity
Systemic-fiscal	AI-crash fallout in munis / banks	AI-crash-insulated CA muni screen; WAL NDFI kill-class + observer channel	muni sleeve screened to the insulated cohort; WAL instrumented
Civilizational	the 7-tail wargame (freeze / war / infra / nuclear...)	tail-wargaming framework: Taleb barbell audit — 3 traps struck (self-dealing art trust, don't-pay-taxes bet, two-way leverage), 3 gaps filled	positive-carry convexity ONLY

The organizing principle that survived the wargame: *positive-carry convexity* — every black-swan long must pass “would I be happy owning it if the swan never comes?” The desk explicitly declined negative-carry insurance (naked puts, tail programs) on net-buyer logic; what it builds instead: - **The convexity backtest verdict (the go/no-go, run honestly):** crash convexity is nearly monopolized by **gold-royalty** (the only vertical that held through the 2020 liquidation); defense/energy/ag are war-convex but crash-beta-worse (a latency trade); and four fashionable “chaos” verticals (cyber, uranium, rare-earth, reshoring) tested **beta-1 or worse and were dropped** — the discipline most thematic products lack. The surviving construction: gold-royalty core + resilient thin-coverage carry (CWCO/PLPC-class) + war/inflation satellite + metal tail. Current book expressions: PHYS 320 sh + resting adds, AAUC, CAML — the real-asset-ballast axis at \$10k vs its \$150k target is the most under-built intended axis in the construction plan. - **Dry powder as the option:** the September \$5M and the fade ladders far below market ARE the long-black-swan — a crash is a windfall to the resting buyer. - **The mundane-tail correction:** $P(\text{death/illness/lawsuit}) \gg P(\text{nuclear war})$; the cost-weighted priorities are boring (estate plan, insurance, custody diversification) and sit outside this doc but inside the household queue. - **The deep reframe:** SignalOS itself is the ultimate antifragile asset — a *capability* (find dislocations, read thin paper, be the patient bidder when others are forced sellers) that appreciates in disorder. The product expression is **Antifragile Carry** (spec v0.1 on file: capacity-capped, carry-not-bleed, verified-not-thematic; the calibration ledger as the un-gameable track record).

15. The unified risk picture (how Parts I–IV interlock)

The TLH sleeve supplies the beta and the tax convexity; the alpha sleeve supplies the anti-rotation construction and the capacity-walled edges; the ensemble supplies the one number (0.45) and the sequence that price the hedge; the disaster book supplies the filter (carry-only convexity) that keeps every layer positive-EV in the base case. One risk, one owner, no layer asked to do another’s job: rotation → construction; beta/AI-tail → the two-layer XND hedge + the tax code; fiscal contagion → screened munis + the WAL instrumentation; the civilizational tail → gold-royalty carry, dry powder, and a desk that gets more valuable in disorder.

Decision queue (principal)

0. Stage (or decline) the designed IGV rotation-hedge envelopes and the real-asset-ballast deployment priority (Part IV).
1. Ratify the two-layer hedge + the conditioned-insurance amendment (or choose a Step-4 menu row).
2. Ratify or decline the dispersion sleeve (Part I §4).
3. Sec-ALPHA-MIGRATION gates into the runbook; gateway-restart + envelope-launchd autonomy calls.
4. BETA activation checklist on approval.

Figures re-derive at landing and at each quarterly re-freeze; nothing stages until the \$5M lands and item 1 is ratified.